Swiss Institute of Bioinformatics

SHARE – CONNECT – UNDERSTAND

# Toward sustainable life sciences

## Privacy-preserving AI and multimodal data integration

**Trần Thượng Vân Du, Vital-IT**

VSOB 2026

elixir SWITZERLAND

# We have never had more molecular and clinical data. Most of them can never be analysed **together**.

In obesity research alone, IMI-SOPHIA had to bring together 16 cohorts, without moving patient-level data.

# Why the potential stays locked



## Privacy

Sensitive data that cannot, and should not, leave its source

Share



## Fragmentation

Data scattered across institutions, formats and domains

Connect



## Interpretability

Models that predict well but explain little about biology

Understand

# One path, three steps

## Share

Privacy-preserving federated analytics and AI/ML

## Connect

Multimodal integration across domains

## Understand

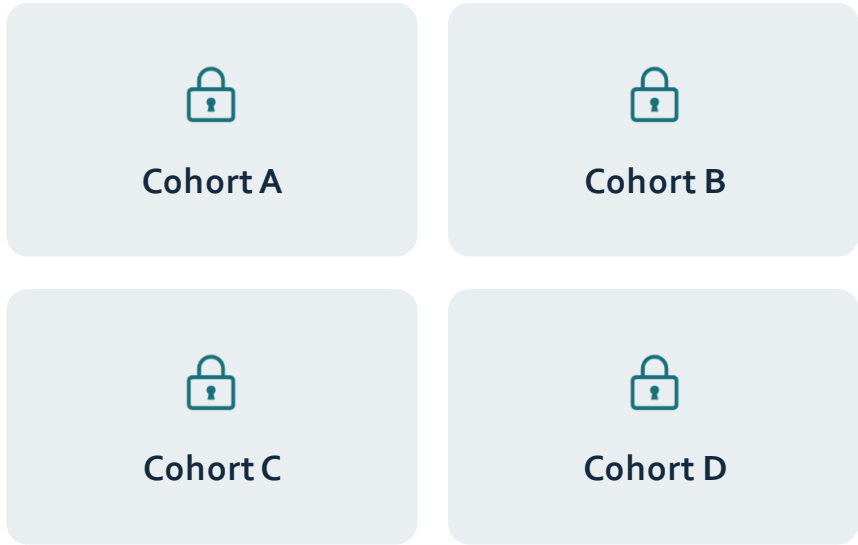
Mechanistic systems biology

# Sensitive data stays in silos

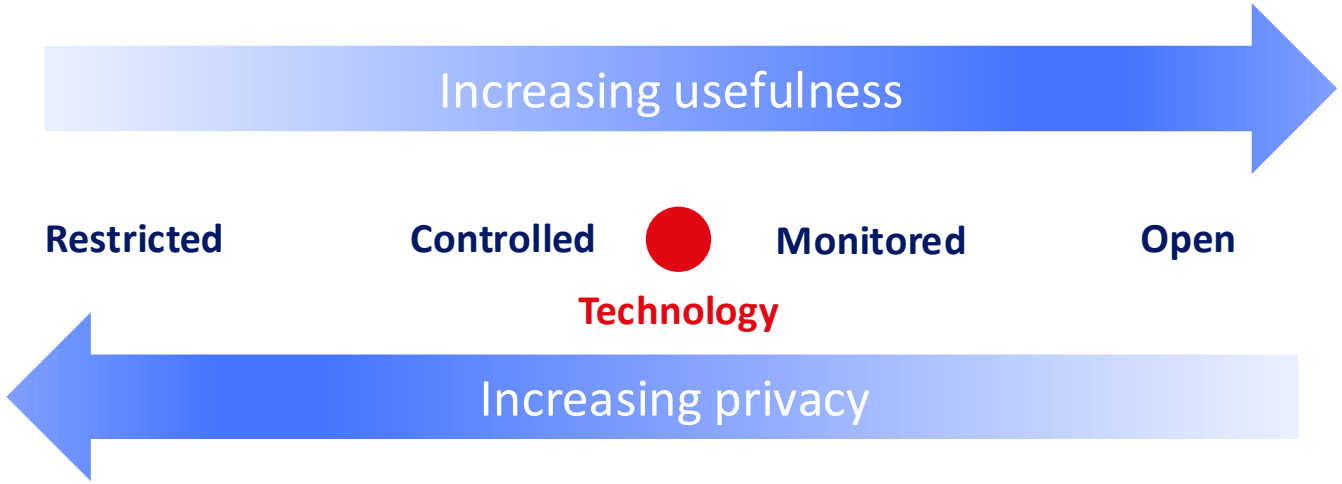
**Too small alone.** Single cohorts lack the power for robust biomarkers.

**Hard to pool.** Central data transfer is legally complex, slow and risky.

**Costly copies.** Every transfer duplicates data and storage.



# Finding the sweet spot between privacy and usefulness



# The analysis travels, the data stays

## Harmonise

16 cohorts: research studies, clinical trials and routine health data

## Analyse locally

Open-source DataSHIELD runs statistics and ML remotely

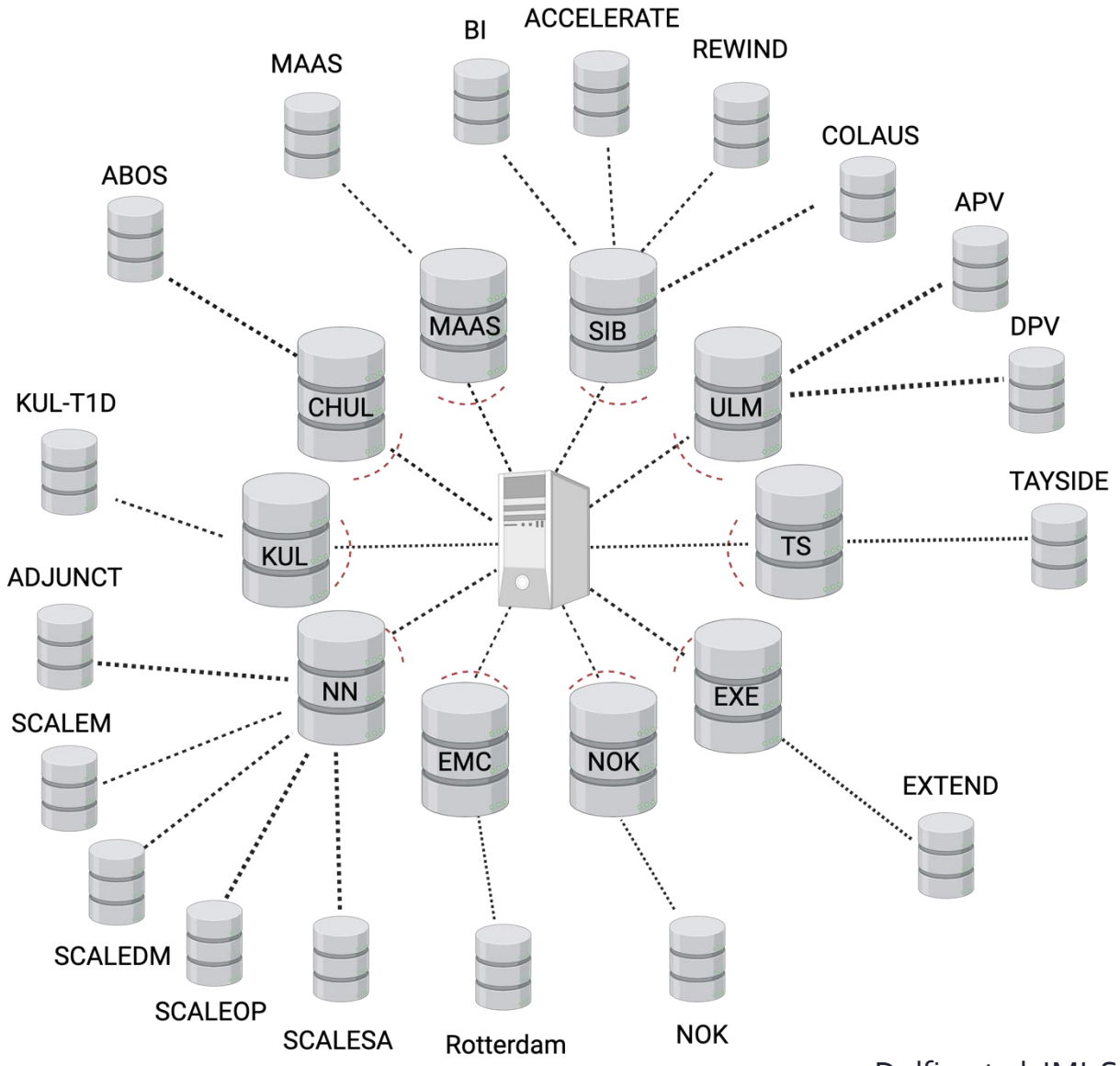
## Aggregate

One central gateway; only non-disclosive results return

**Individual-level data never leaves its source.**

Share

# The SOPHIA Federated Database




**16** harmonised (OMOP), federated obesity cohorts hosted on **10** servers in **8** countries:

- *Netherlands (Rotterdam, Maastricht, Amsterdam)*
- *Belgium (Leuven)*
- *Denmark (Copenhagen)*
- *Scotland (Tayside)*
- *England (Exeter)*
- *France (Lille)*
- *Switzerland (Lausanne)*
- *Germany (Ulm)*

Clinical data for >900K patients

Delfin et al. IMI-SOPHIA federated database (2024)

Share

SOPHIA Stratification of Obese Phenotypes to Optimize Future Obesity Therapy

SIB

# Naming something correctly is not always obvious!



# Data harmonisation

## Data cleaning

| GENDER Original | GENDER Clean |
|---|---|
| Male | M |
| M | M |
| m | M |
| Female | F |
| F | F |
| f | F |
| unknown | |
| na | |

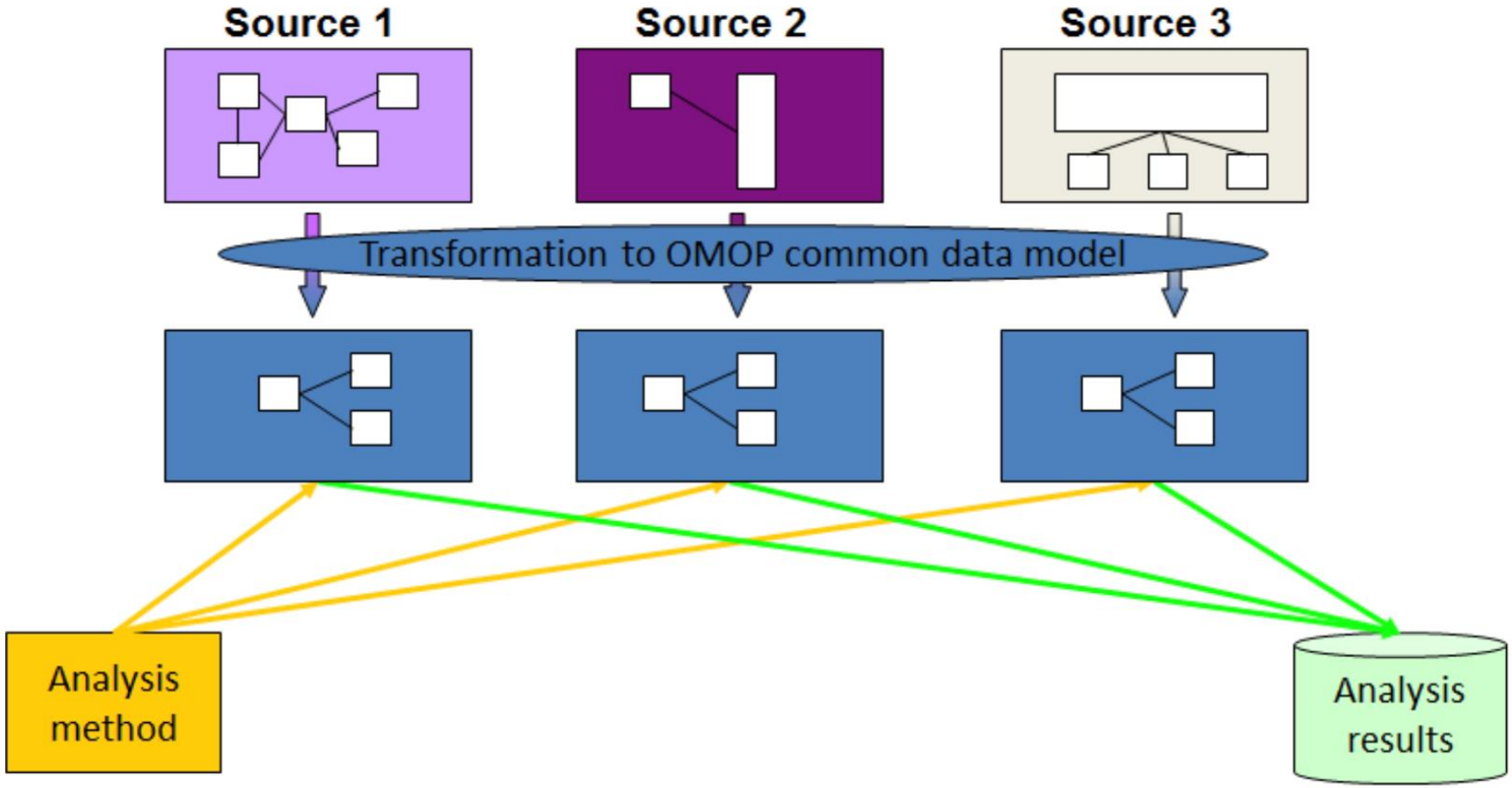
## Standardization or **Harmonisation** or Mapping

| GENDER Clean | Standard Concept | Standard Concept ID |
|---|---|---|
| M | gender_concept | 8507 |
| F | gender_concept | 8532 |

**Identifier we will use (e.g. in a database)**

Mark Ibberson (SIB)

# Standard data model




https://www.ohdsi.org/data-standardization/the-common-data-model/

Share

# Federated analysis



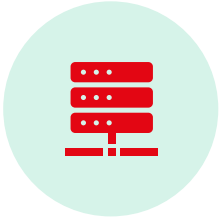
Perform the analysis interactively on multiple remote datasets



Take the analysis to the data rather than bring the data to the analysis tool



Sensitive data never leave their host institution



Only non-disclosive data are shared between servers and client

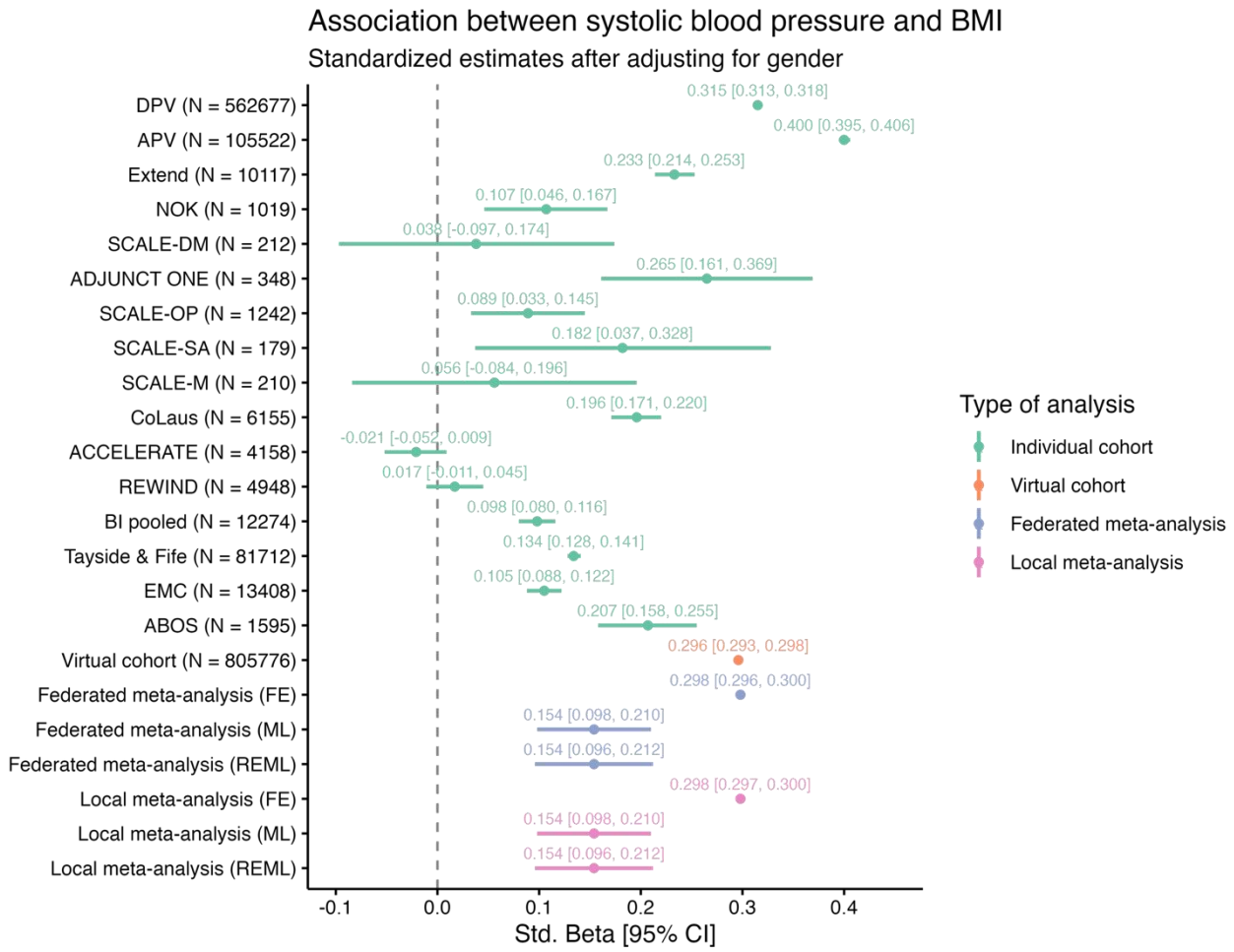
# Principles of secured communication

- Sensitive data should not be identified
  - directly: hacking the transfer
  - indirectly: hacking the transfer + decrypting/solving

- Assume encryption can be broken: share only non-disclosive results

(encryption still helps discourage hackers!)

# Federated analysis matches meta-analysis




**16**

studies pooled in one federated model

**1**

central gateway, accessed from R

**0**

patient-level records seen by analysts

Delfin et al. IMI-SOPHIA federated database (2024)

Share

SOPHIA
Stratification of Obese Phenotypes to Optimize Future Obesity Therapy

SIB

# Federation as a sustainability strategy

### Privacy by design

Data owners keep control; trust makes collaboration possible

### No redundant copies

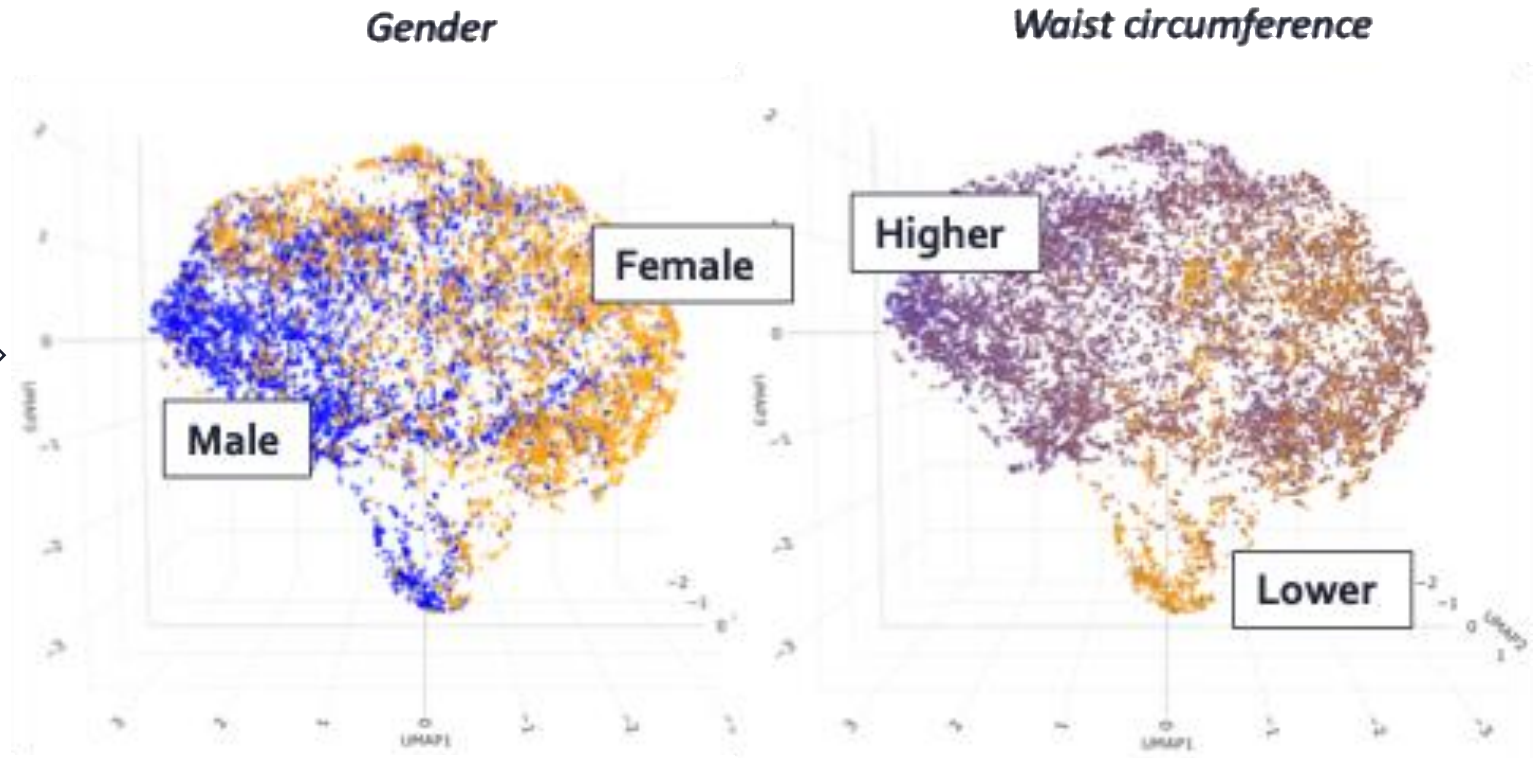
Less duplication, transfer and storage of the same data

### Long-term reuse

Harmonised data stays ready for the next study

# Cross-cohort Federated UMAP using clinical variables

Body weight
Body height
Cholesterol
Diastolic blood pressure
Systolic blood pressure
Triglycerides
Waist circumference
Age (>65)



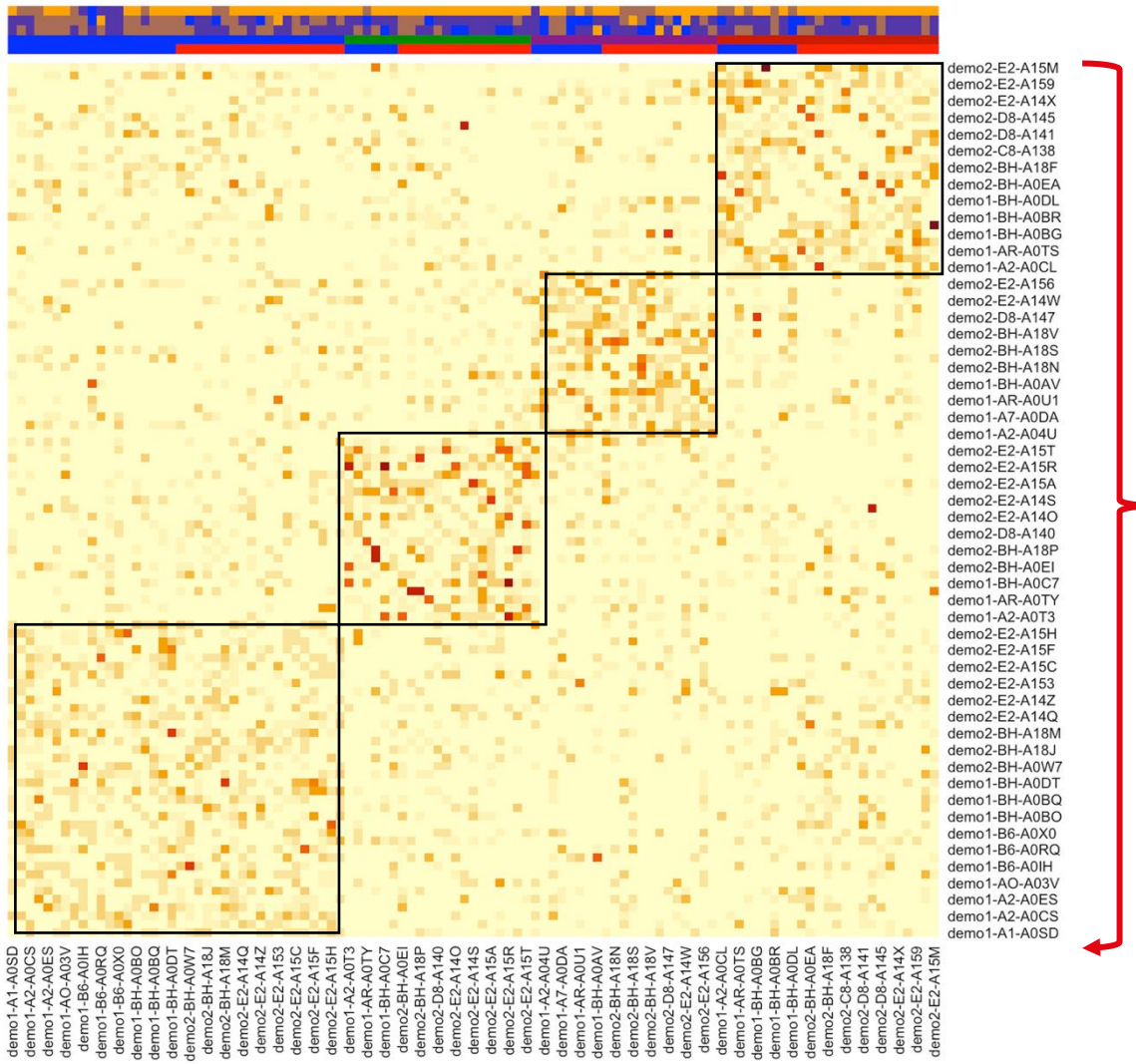

***Cohorts:***

- *CoLaus (N=1014)*
- *REWIND (N=1060)*
- *ACCELERATE (N=3127)*
- *Rotterdam (N=3228)*

Share

SOPHIA
Stratification of Obese Phenotypes to Optimize Future Obesity Therapy
SIB

# Federated Similarity Network Fusion on Breast Cancer




*Horizontal integration of multi-omics data across federated nodes:*

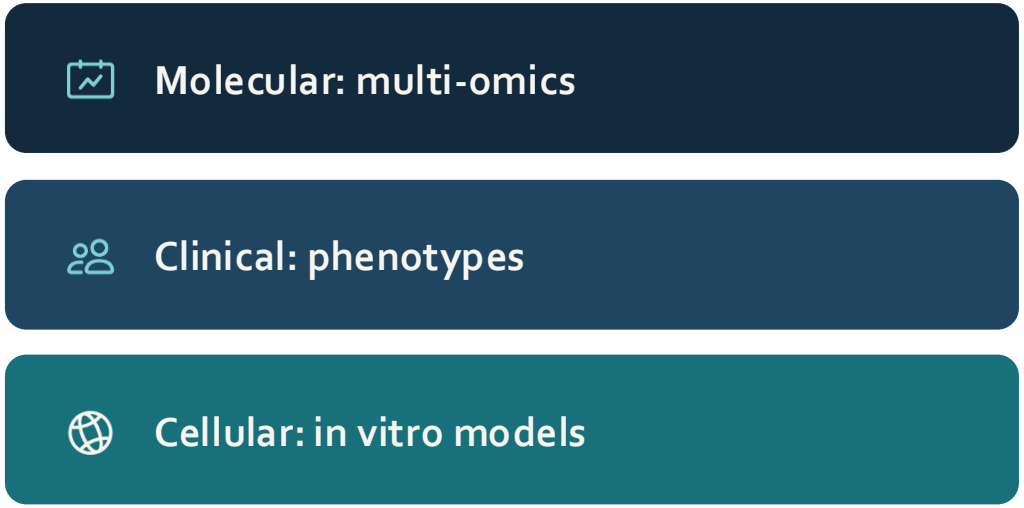
**Sample-level data on 2 federated nodes**

- *Methylomics*
- *Gene expression*
- *miRNAomics*

Data from https://doi.org/10.1038/nmeth.2810

Share

# Integrating data across scales



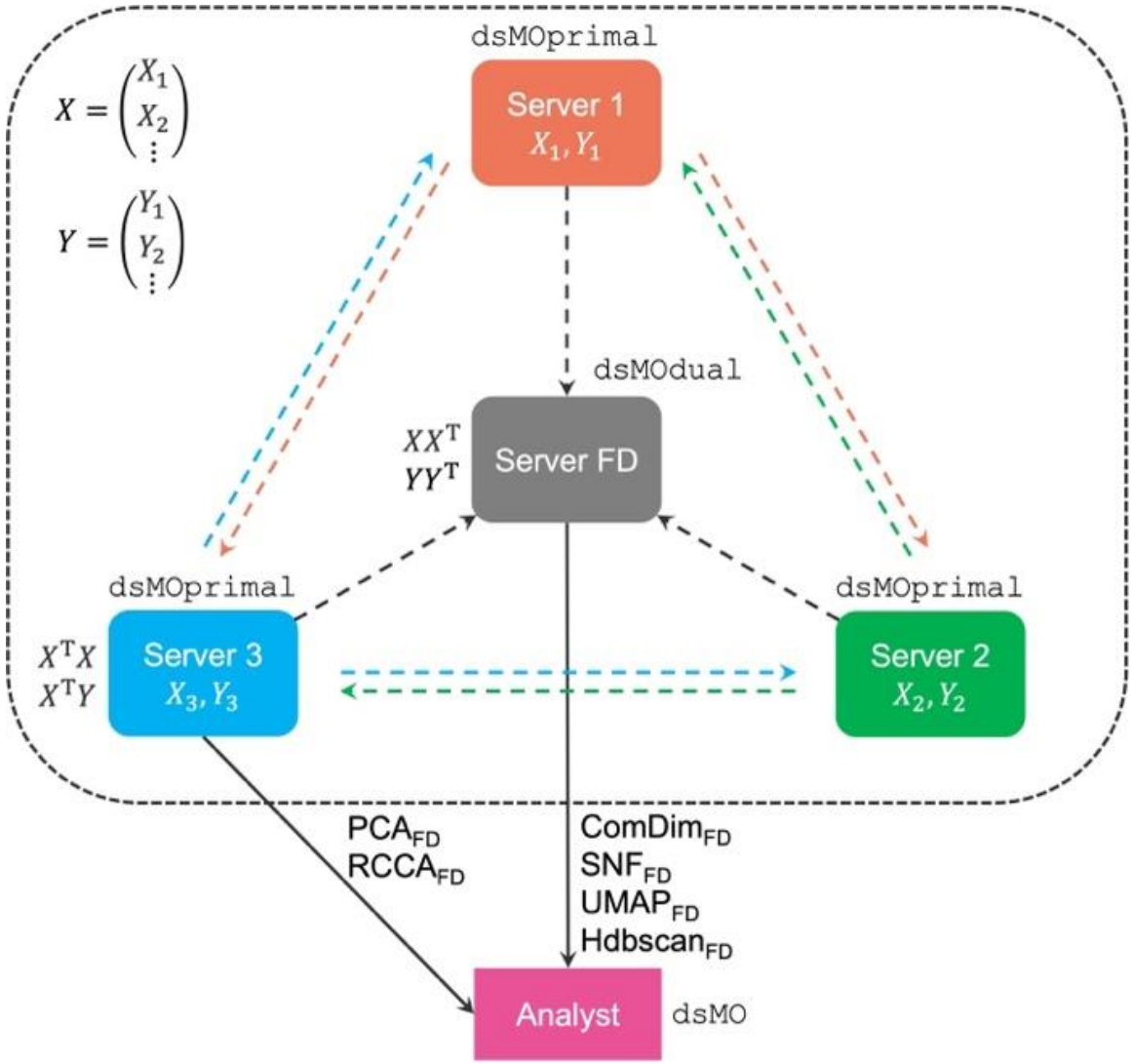

## The challenges

Heterogeneous scales and formats

Missing modalities per sample

Noise and batch effects

# A framework for federated analysis of multimodal data




Tran et al. dsMO R package suit (2018)

- Principal Component Analysis (PCA)
- Regularized Canonical Correlation Analysis (RCCA)
- Common components and specific weights analysis (ComDim)
- Similarity Network Fusion (SNF)
- Uniform Manifold Approximation and Projection (UMAP)
- Hierarchical Density-Based Spatial Clustering (HDBSCAN)

Reimplementing analysis methods with $XX^T, X^TX$ input

# Multi-omics in type 2 diabetes

IMI-RHAPSODY

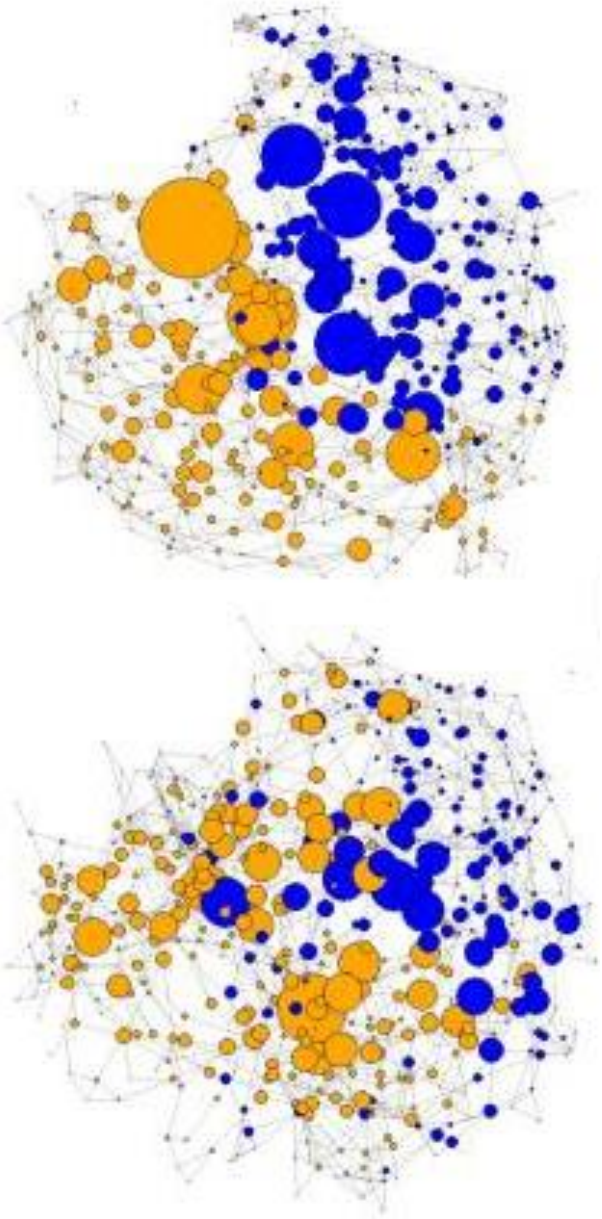
## Subgroups that predict glycaemic deterioration

1,134 people with T2D (DCS, GoDARTS)

2 omics: circulating plasma lipidomics & proteomics

Similarity Network Fusion

**2 subgroups of patients**, differed in terms of glycaemic deterioration, insulin sensitivity and secretion; driven by triacylglycerols, sphingomyelin, testican-1, IL-18 receptor



Li et al. T2D patient stratification (2024)

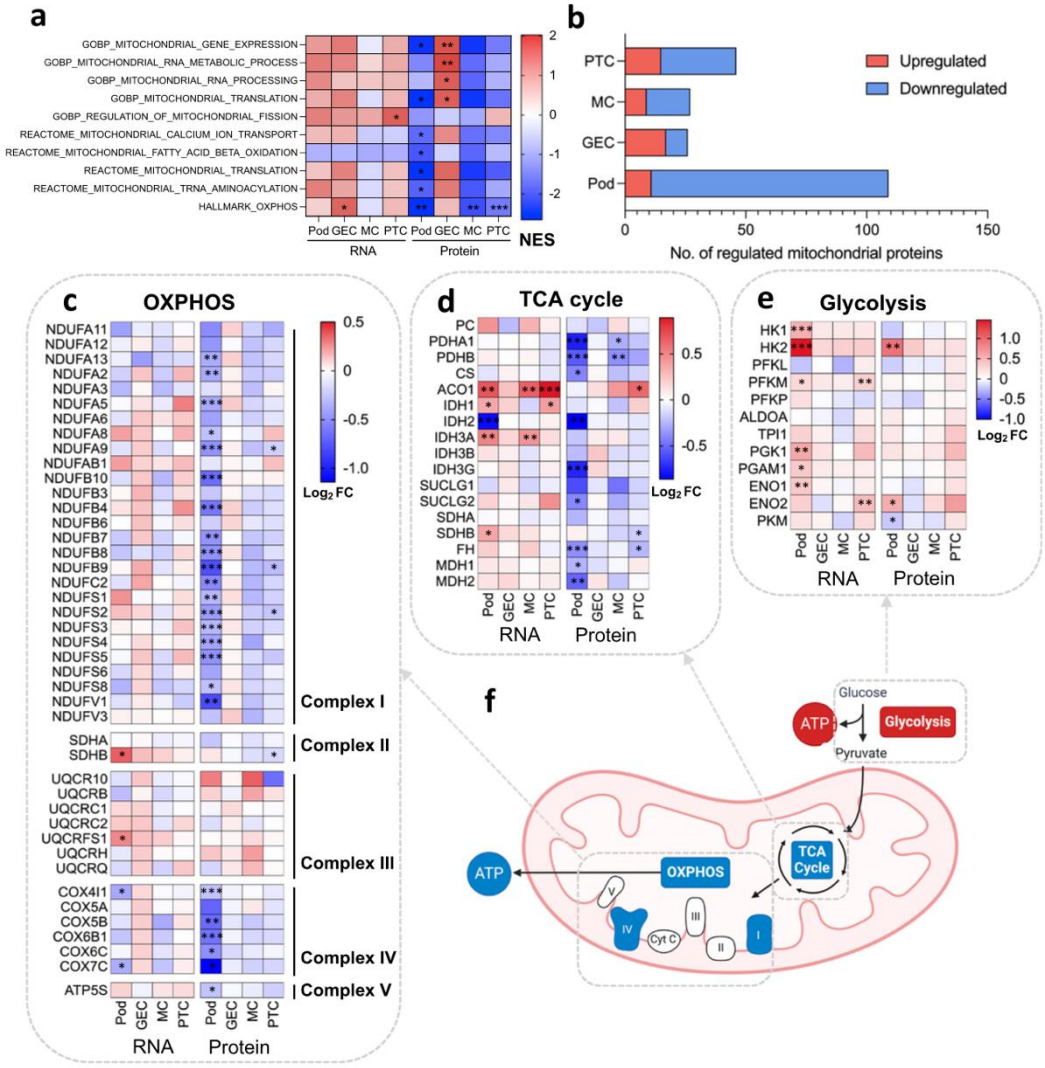
# Multi-omics in diabetic kidney diseases

IMI BEAt-DKD

## Insulin resistance across kidney cell types

4 human cell types: podocytes, glomerular endothelial, mesangial, proximal tubular
2 omics: transcriptomics + proteomics

**Common and cell-type-specific** mechanisms, confirmed in early- and late-stage DKD biopsies




Lay et al. Diabetic Kidney Disease mechanism (2024)

# From prediction to mechanism

PREDICTIVE MODEL

## “Who is at risk?”

Accurate scores, limited biological explanation

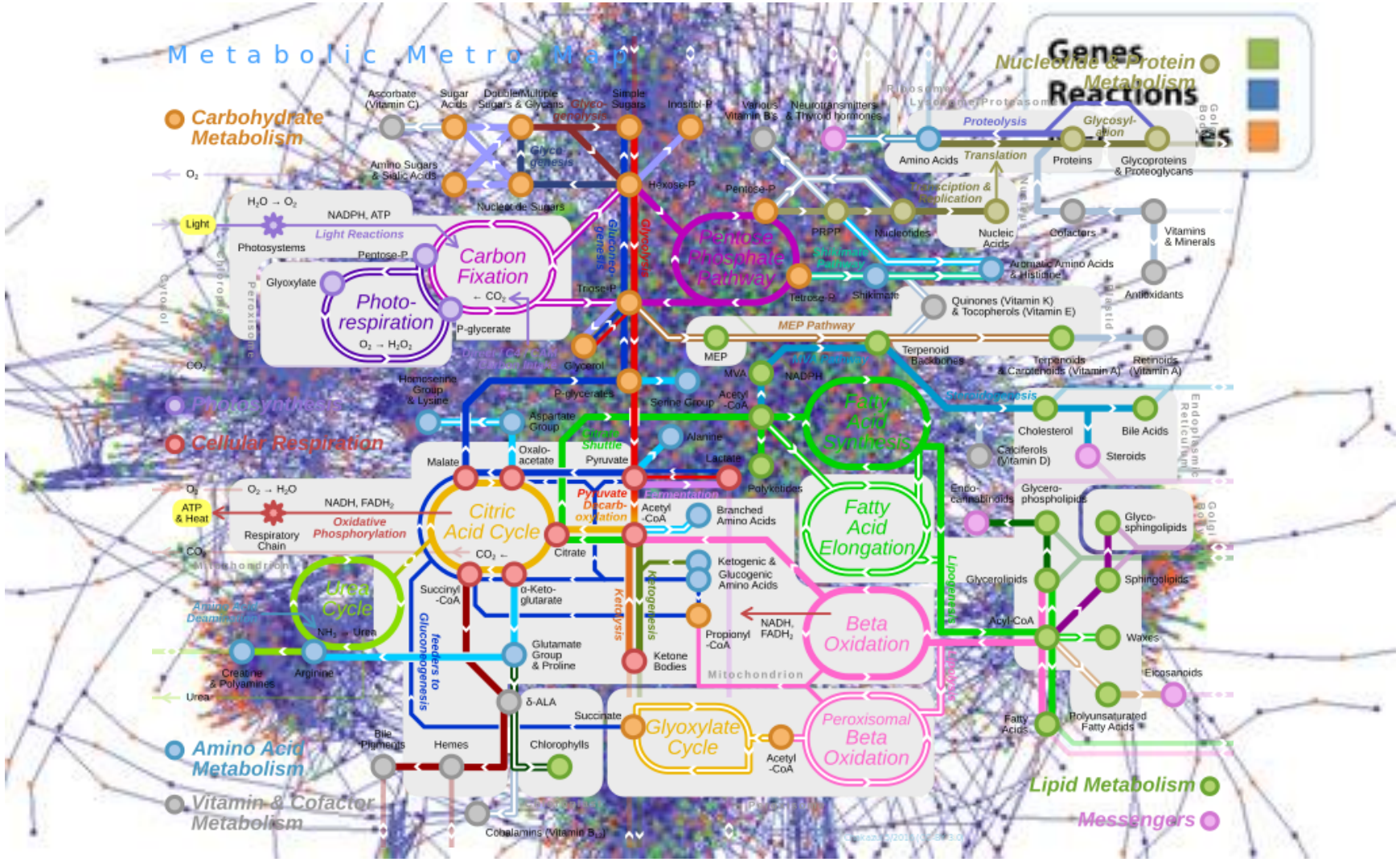


MECHANISTIC MODEL

## “Which pathways drive it?”

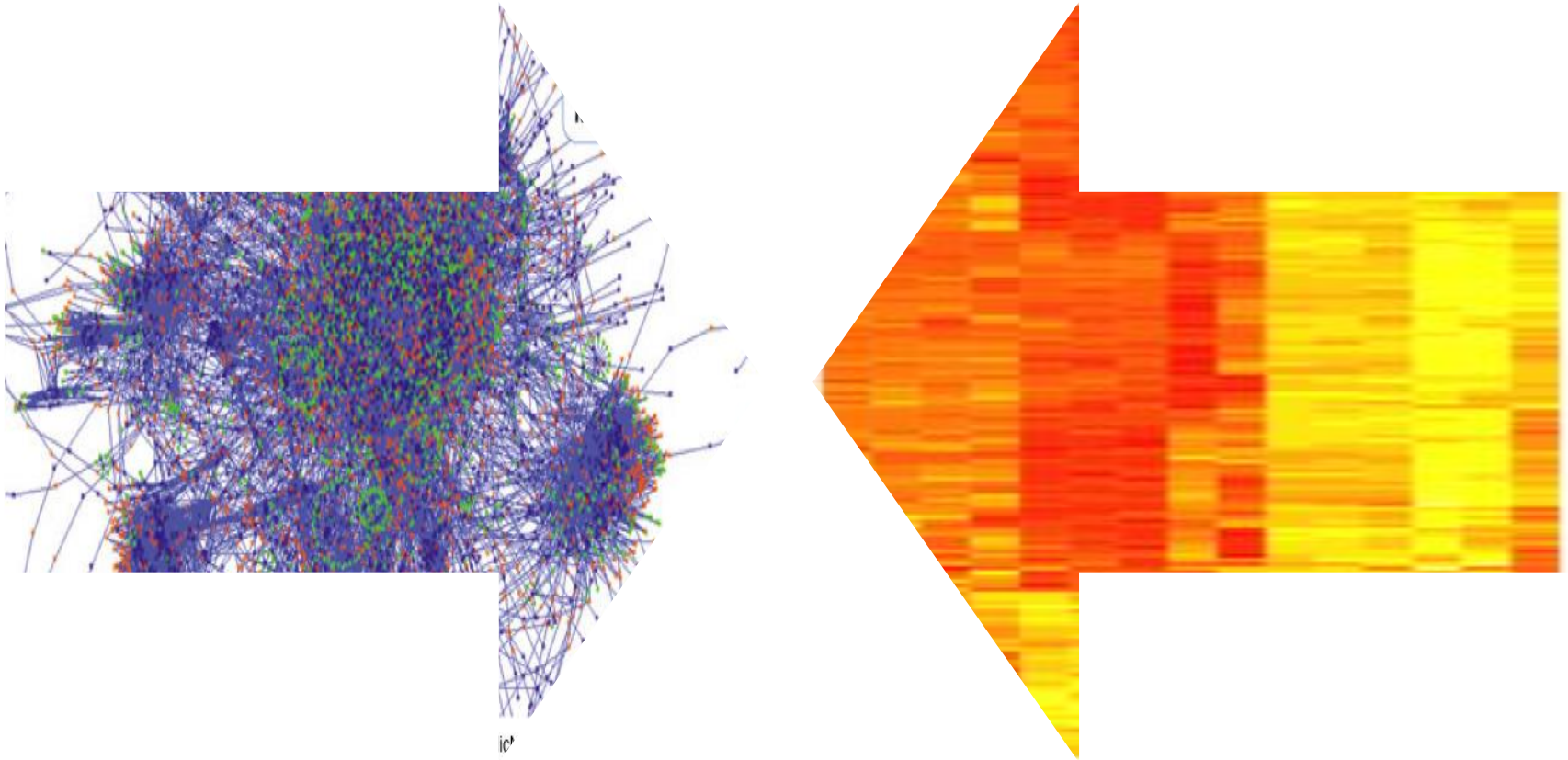
Testable hypotheses and targets for intervention

# Transcriptomics on metabolic networks

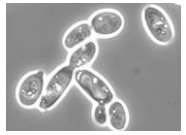


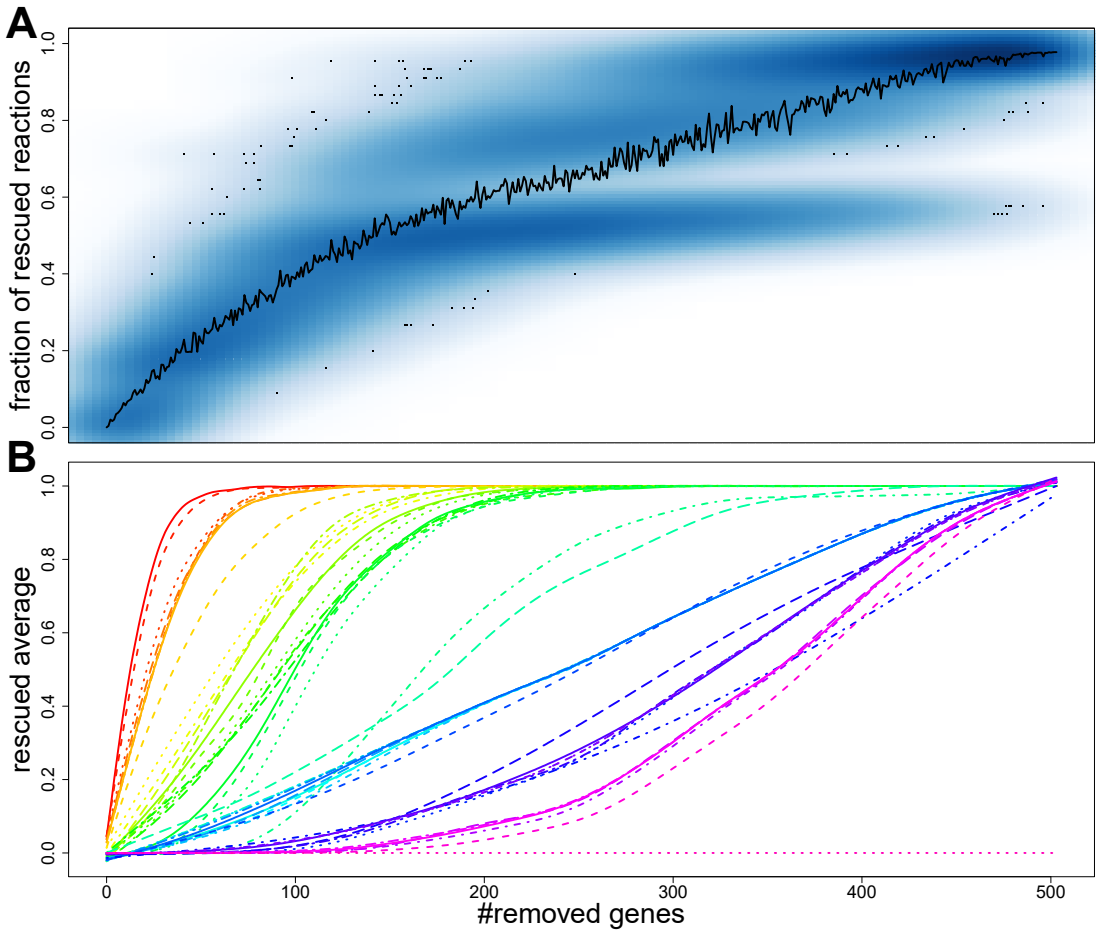

http://www.genome.jp/dbget-bin/show_pathway?map01100 https://commons.wikimedia.org/wiki/File:Metabolic_Network_Map.svg

# Transcriptomics on metabolic networks

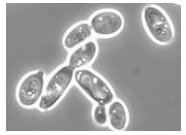


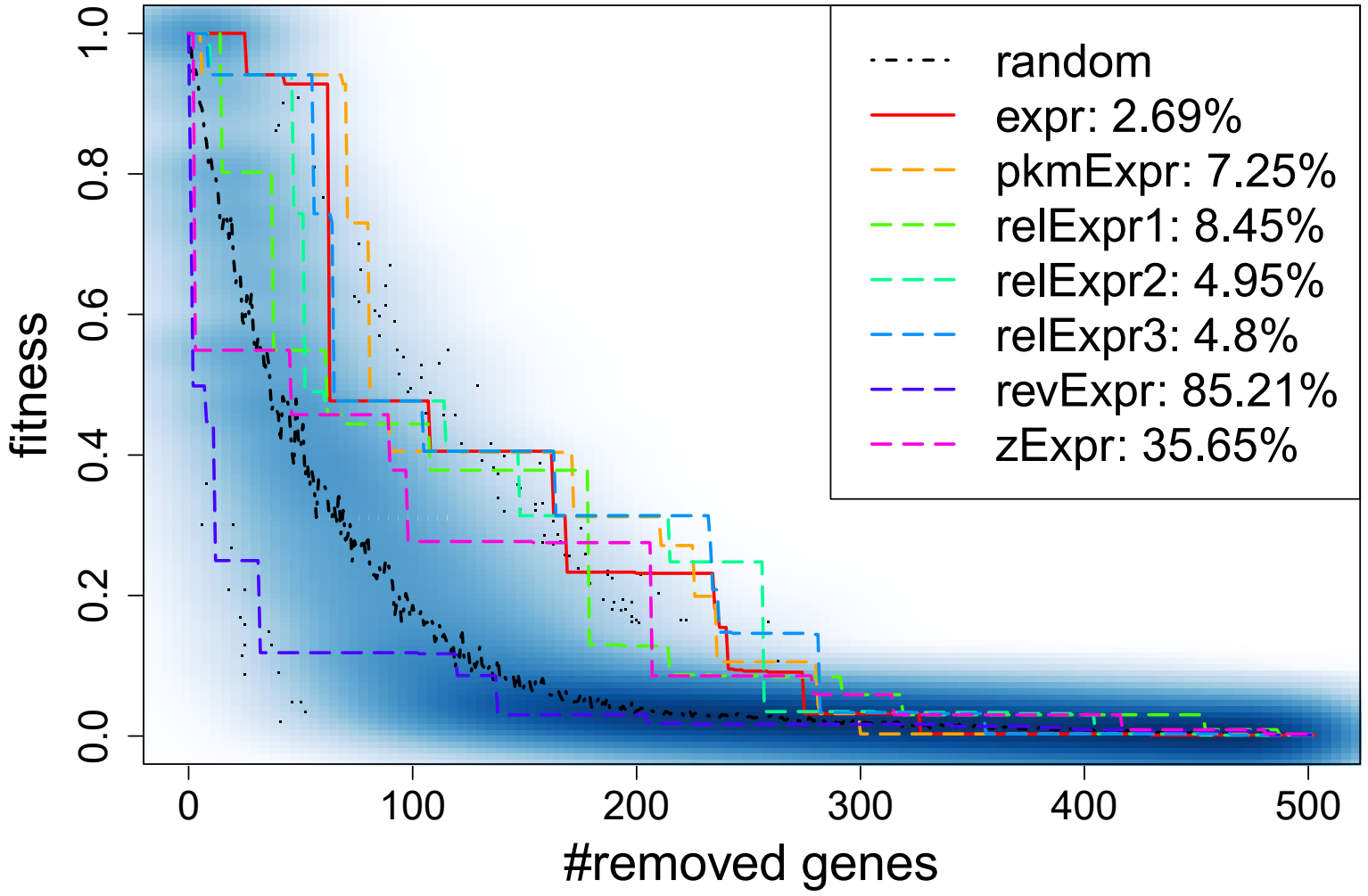
Tran et al. metaboGSE (2018)




A
fraction of rescued reactions
0.0
0.2
0.4
0.6
0.8
1.0
B
rescued average
0.0
0.2
0.4
0.6
0.8
1.0
0
100
200
300
400
500
#removed genes
ergosterol
5alpha−cholesta−8,24−dien−3beta−c
phosphatidyl 1D myo inositol
triglyceride
phosphatidylserine
Phosphatidylcholine
phosphatidylethanolamine
Mannan
alpha,alpha−trehalose
1 3 beta D Glucan
CMP
ATP
a glycogen
asparagine
dCMP
glutamine
dTMP
dGMP
UMP
ADP
GMP
H(+)
dAMP
histidine
valine
threonine
isoleucine
cysteine
tyrosine
tryptophan
phenylalanine
phosphate
proline
sulfate
AMP
aspartate
methionine
arginine
lysine
alanine
serine
leucine
glycine
glutamate
H2O

# RNA-seq: Optimal ranking for gene removal?

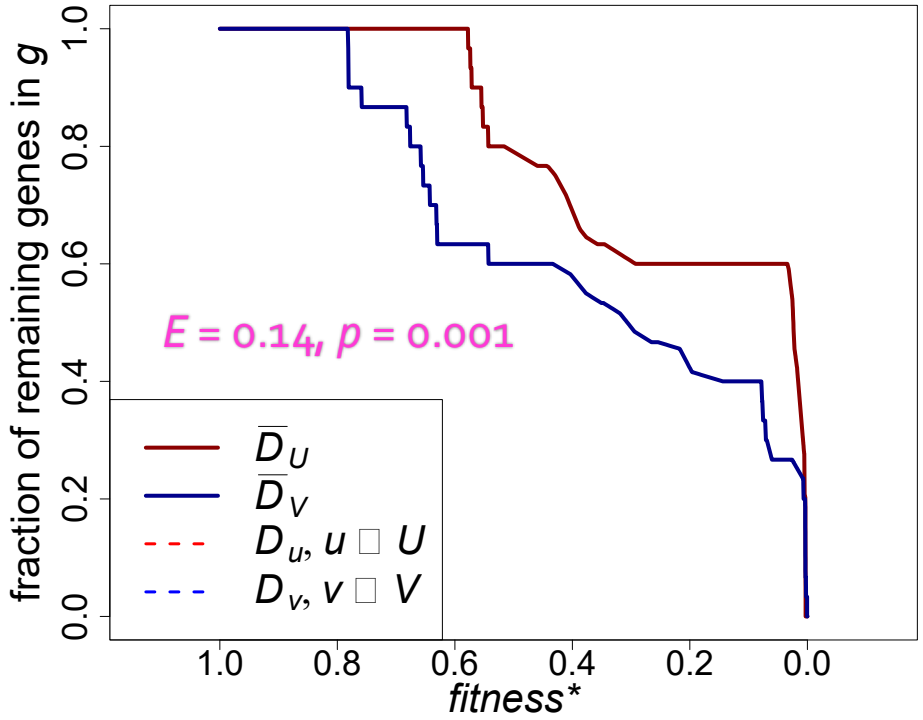




# Gene set enrichment with metaboGSE

$$\bar{D}_U(x) = \langle D_u(x) \rangle_{u \in U}$$

$$x = fitness^* = (1 - i/k).fitness$$




Area between the two average curves

$$E = \bar{A}_{U,V} = \int_0^1 |\bar{D}_U(x) - \bar{D}_V(x)| \mathrm{d}x$$

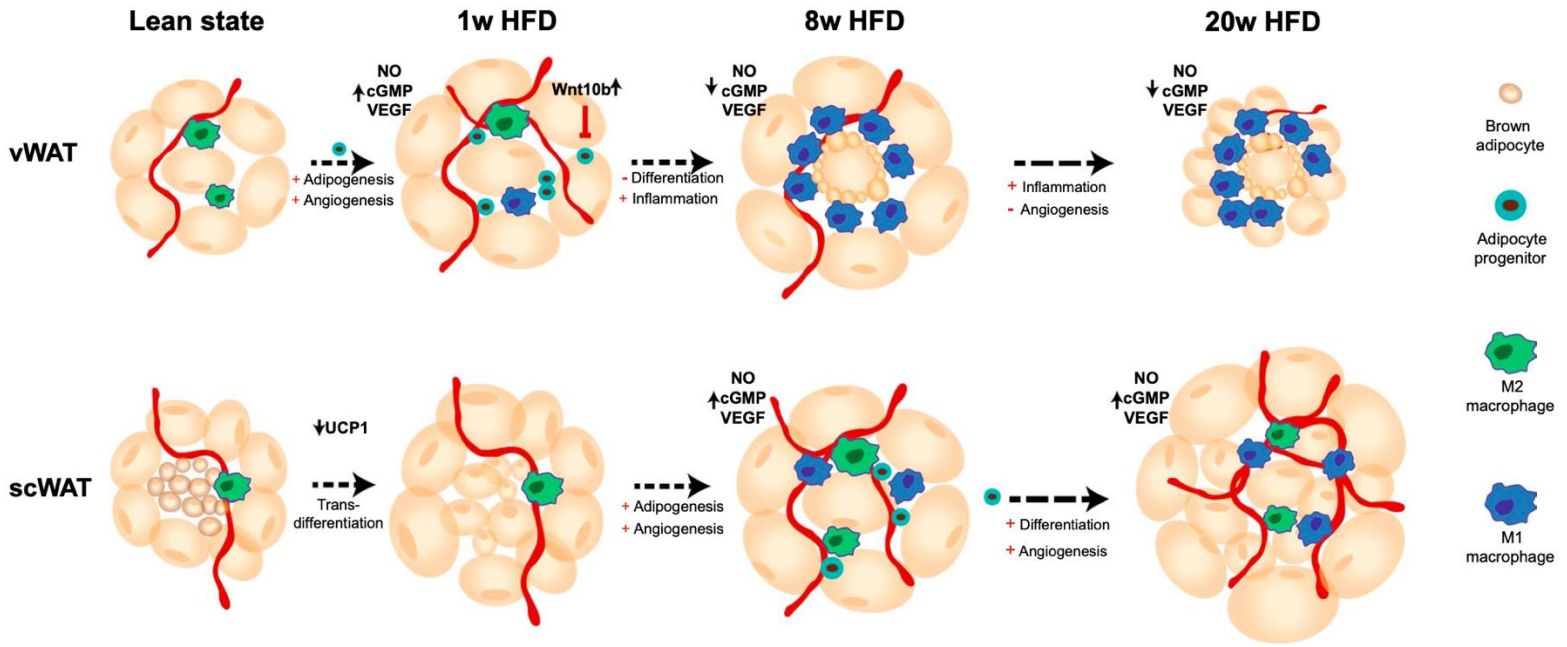
Permutation test on the sample labels to generate a random distribution for *E*

$$p = \frac{\sum_E \mathbb{I}(E \geq E_{obs}) + 1}{\|E\| + 1}$$

$$\Delta_{U,V} = AUC_U - AUC_V = \int_0^1 \left(\bar{D}_U(x) - \bar{D}_V(x)\right) \mathrm{d}x$$

# Drivers of obesity-related inflammation

Mice on high-fat diet: transcriptomics and ChIP-seq analysed with metaboGSE on metabolic networks




Caputo et al. (2020)

**Visceral fat:** precursors proliferate, but anti-adipogenic signals block their differentiation → inflammation

**Subcutaneous fat:** beige adipocytes store fatty acids; angiogenic, hyperplastic growth → expansion without inflammation

# Bringing together

## Share

Analyse sensitive data where it lives

## Connect

Combine multiple views

## Understand

Turn predictions into mechanisms

Toward sustainable life sciences:

privacy-preserving, collaborative, and interpretable.

# Where AI helps in our workflow

## Harmonise

Suggest OMOP concept mappings, flag unit and code mismatches, draft ETL scripts

## Code

Draft, explain and test DataSHIELD and R/Python analyses; write documentation

## Choose methods

Explain when SNF, UMAP or HDBSCAN fit the data and what each parameter does

## Interpret

Summarise literature on markers and pathways; turn results into testable hypotheses

## Annotate

Map genes, proteins and lipids to pathways and functions; check identifiers across databases

## Federated ML

Train models where the data lives: the privacy-preserving AI

# Where to be careful

## Data & privacy

**Never paste sensitive data**
into public chatbots: the analysis travels, not the data

**Models can leak**
via membership inference or model inversion, even when federated

**Bias stays**
federation pools cohorts but does not make them representative

## Correctness

**Hallucinations**
invented references, gene functions and package functions

**Plausible but wrong code**
silent errors in normalisation, joins, gene IDs

**Statistics at speed**
leakage, overfitting, uncorrected tests, over-read UMAP clusters

## Responsibility

**Prediction ≠ mechanism**
AI output is a hypothesis, not an explanation

**Reproducibility**
version code; record models, prompts and seeds; disclose AI use

**Environmental cost**
use the smallest tool that works; reuse data and models

# Treat AI like a fast junior colleague

**Verify**

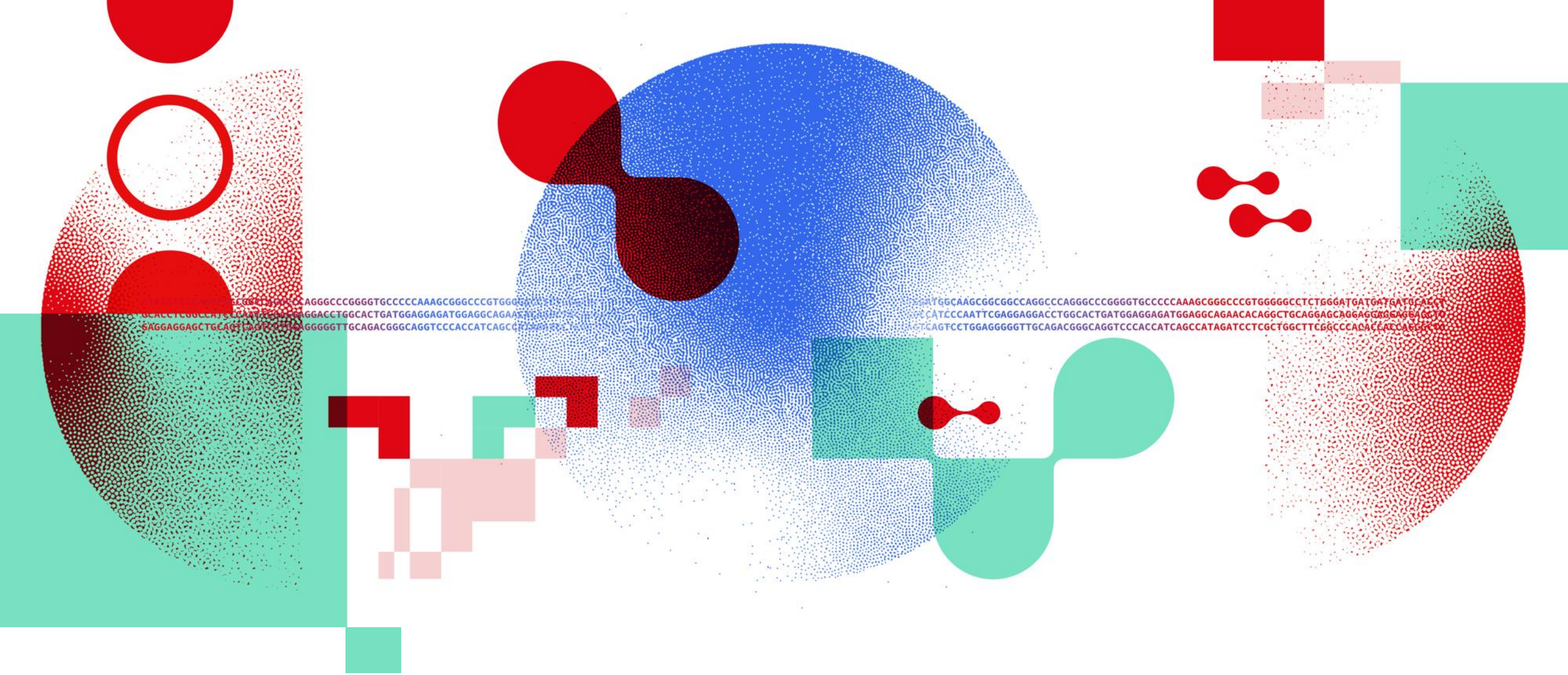
every output element

**Protect**

keep sensitive data where they live

**Document**

code, model versions, prompts, AI use



# Thank you!

elixir SWITZERLAND

SIB